

AIMS
Airbus Material Specification

Copper alloy (CuNi15Sn8)
Solution treated and spinodally hardened (TX00)
Rm $\geq$ 900 MPa

Wrought round or hollow Bar
Diameter $15 \text{ mm} \leq D \leq 108 \text{ mm}$ or
Thickness $7 \text{ mm} \leq a \leq 30 \text{ mm}$ with $50 \text{ mm} \leq D \leq 200 \text{ mm}$

Material Specification

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1 Scope

This Material Specification defines the requirements of wrought round or hollow bars with diameter $15\text{ mm} \leq D \leq 100\text{ mm}$ or thickness $7\text{ mm} \leq a \leq 30\text{ mm}$ with $D \leq 200\text{ mm}$ in copper alloy (CuNi15Sn8), $R_m \geq 900\text{ MPa}$, for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

Typical application is pylon to wing attachment.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ABS5152	Aerospace series - Hollow bar - Copper alloy (Cu15Ni8Sn) - Thickness $7\text{ mm} \leq a \leq 30\text{ mm}$ with $50\text{ mm} \leq D \leq 200\text{ mm}$
ABS5849	Aerospace series - Round bar - Copper alloy (CuNi15Sn8) - Diameter $15,0\text{ mm} \leq D \leq 108,0\text{ mm}$
AIMS02-04-000	Technical specification – Copper alloy – Wrought products Plate, sheet, strip, bar, hollow bar and section
ASTMB601	Standard Classification for Temper Designations for Copper and Copper Alloys - Wrought and Cast
EN2032-02	Metallic materials - part 02 -- coding of metallurgical condition in delivery condition
EN4500-1	Aerospace series – Metallic materials – Rules for drafting and presentation of material standards – Part 1:General rules.

4 Technical requirements

4.1 General

Specification values are minimum except where stated

4.2 Specific

The data requirements given in the tables are according to the relevant EN4500-1.

5 Designation

Refer to ABS5152 and ABS5849.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS02-04-000.

1	Material designation				CuNi15Sn8							
2	Chemical composition%		Element	Ni	Sn	Fe	Zn	Mn	Nb	Mg	Pb	Cu
			Mini	14,5	7,5	-	-	-	-	-	-	The rest
			Maxi	15,5	8,5	0,50	0,50	0,30	0,10	0,15	0,02	
3	Method of melting				-							
4	Form Method of production Limit dimension (mm)				Round or hollow bar Wrought products $15\text{mm} \leq D \leq 108 \text{ mm}$ or $7\text{mm} \leq a \leq 30\text{mm}$ with $D \leq 200\text{mm}$							
5	Technical specification				AIMS02-04-000							
6	Delivery condition and heat treatment				Solution treated and Spinodally Hardened TX00 ¹⁾ For details see relevant IPS ²⁾							
7	Use condition and heat treatment				As delivered							
8	Sample Test piece Heat treatment				Solution treated and Spinodally Hardened							
9	Dimension concerned	a or D	mm	$D \leq 108 \text{ mm}$ or $7\text{mm} \leq a \leq 30\text{mm}$ with $50\text{mm} \leq D \leq 184\text{mm}$				$7\text{mm} \leq a \leq 30\text{mm}$ with $184\text{mm} < D \leq 200\text{mm}$				
10	Thickness of cladding on each face		%	-								
11	Direction of test piece				L							
12	Tensile	Temperature	θ	°C	Room temperature							
13		Proof stress	R _{p0,2}	MPa	≥ 760							
14		Strength	R _m	MPa	≥ 900							
15		Elongation	A	%	≥ 10			≥ 8				
16		Reduction of area	Z	%	-							
17	Hardness		HRC	-	$30 \leq \text{HRC} \leq 36$							
19	Bending		K	-	-							
20	Impact strength		K	-	-							
21	Creep	Temperatur e	θ	°C	-							
22		Time		h	-							
23		Stress	σ _a	MPa	-							
24		Elongation	a	%	-							
25		Rupture stress	σ _R	MPa	-							
26		Elongation at rupture	A	%	-							
27	Notes (see line 98)				1) 2)							

30	Microstructure	-	AIMS02-04-000	
		7	See relevant IPS ²⁾	
34	Grain Size	-	AIMS 02-04-000	
		7	Average grain size $G \geq 5$	
44	External defects	-	See AIMS 02-04-000	
85	Bearing	-	See AIMS 02-04-000	
		3	Direction L	
		7	e/d=1,5	Fbru ≥ 1400 MPa
98	Notes	-	1) See ASTM B601. 2) These tests have no specific requirements. They have to be fixed and contractualized through qualification results in the IPS.	
100	Qualification	-	Frequency of qualification testing is included in AIMS02-04-000.	

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 04/04		New standard
2 05/11		Introduction of round bars according to ABS5849. Grain size: Average change G=7 to G=5 and ALA G=5 to G=2 (supplier request) Added ASTMB 601 temper TX00 Modification of dimensions limits.
3 04/13		Line 34, Grain size : following sentence has been deleted “Occasional grains with maximum size G > 2”